

This is a review submitted to Mathematical Reviews/MathSciNet.

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Mathematical Reviews/MathSciNet Reviewer Number: 82361

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Title: A kernel test for causal association via noise contrastive backdoor adjustment.

MR Number: MR4758131

Primary classification:

Secondary classification(s):

Review text:

sectsty graphicx amsmath,amssymb

This article proposes a non-parametric approach to testing for the presence of a causal effect by extending the work of Gretton et al. on the Hilbert-Schmidt Independence Criterion (HSIC) using g-computation principles. The specific contributions are as follows:

- **Introducing Backdoor-HSIC:** Backdoor-HSIC is a novel extension of HSIC derived using importance-weighted covariance operators. The convergence properties of the corresponding estimators in causal settings have also been established. Also developed is a new optimization strategy to improve the effective sample size.
- **Novel Permutation Strategy:** This work provided a permutation strategy for backdoor-HSIC that ensures a permutation test with theoretically correct size for arbitrary treatment types.
- **Empirical Evaluation:** Demonstrating that backdoor-HSIC achieves correct size and exhibits strong power across various treatment types and with a large number of confounders when testing the do-null hypothesis (see the definition in the paper).
- **Ablation Studies and Limitations:** The article also conducted ablation studies to analyze backdoor-HSIC and characterized the circumstances under which it may become invalid.